PIONEER CDJ-1000MK2 → ESP32-S3

New life for a **club legend**

Gutting a 2003 CD deck and giving it a modern brain — a native **USB-MIDI controller** for Traktor Pro 4.

• architecture v0.4 | v0.1 scope locked | GPL-3.0

Why resurrect 23-year-old hardware?

Because nothing made today feels like this. The CDJ-1000MK2 was the **club-standard deck of its era** — a cast-metal chassis and a mechanical jog wheel engineered for nightly abuse.

THE FEEL

A real platter

A heavy, bearing-mounted jog with vinyl-grade touch response you can't buy in a cheap MIDI controller.

THE BUILD

Built like a tank

Aluminium top plate, satisfying switches, illuminated rings — industrial design that has aged beautifully.

THE STORY

Salvage, not landfill

A dead-or-obsolete icon becomes a one-of-a-kind instrument — and keeps e-waste out of the bin.

2003
YEAR OF THE MK2

100 mm
PITCH-FADER THROW

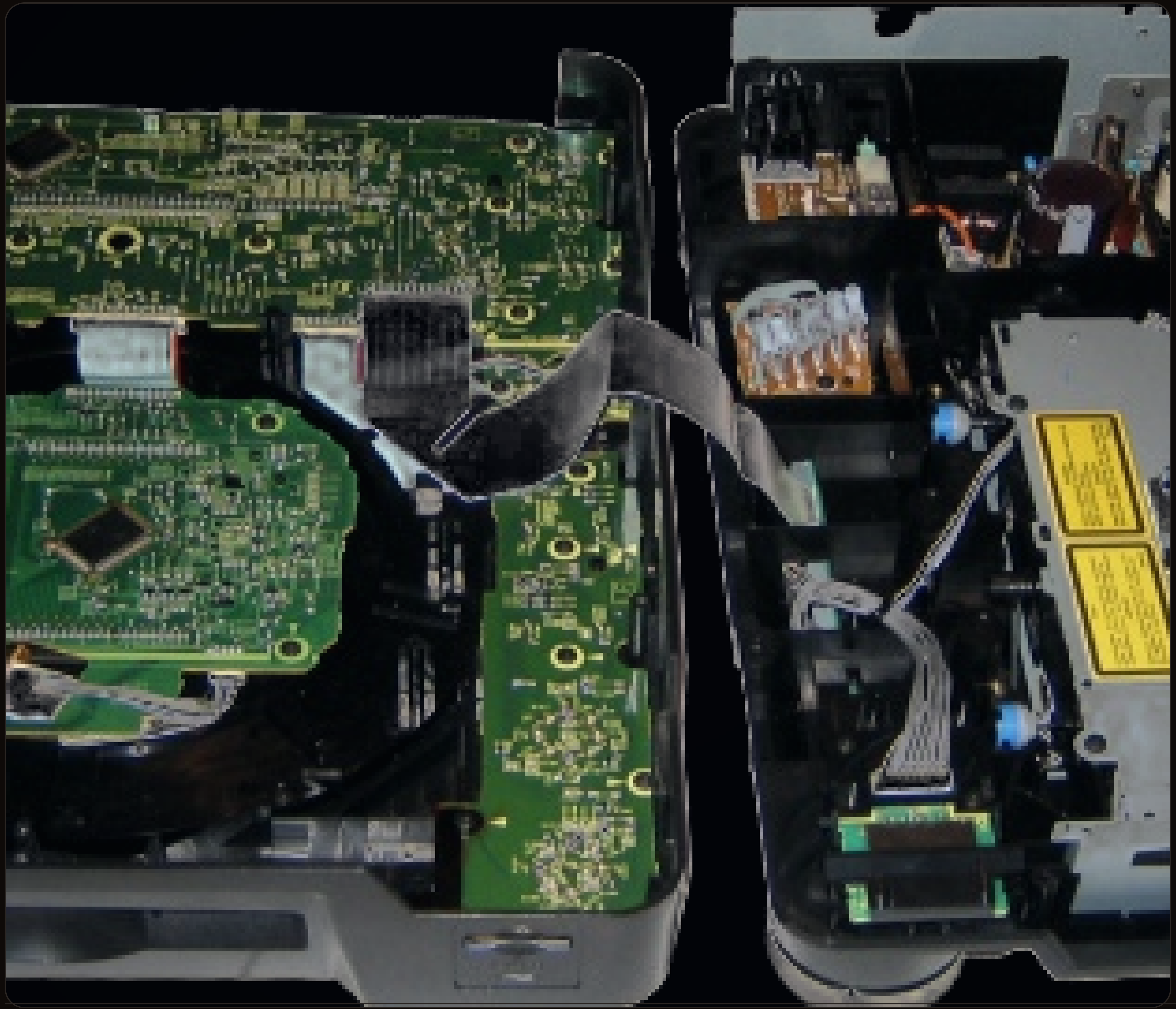
~135 PPR
JOG ENCODER RESOLUTION

It speaks CDs, not computers.

The CDJ-1000MK2 has **no USB and no MIDI**. Every knob, button and jog gesture is locked inside a 2003 mainboard that only knows how to drive a CD transport.

To make it talk to Traktor, the **brain has to go** — but the body is worth keeping.

Keep the body. Gut the brain.



▲ KEEP

The chassis & every control

Jog assembly, 100 mm pitch fader, all button PCBs — including the MK2 Hot Cue A/B/C + Hot Loop — pots, encoders and the illuminated CUE/PLAY rings.

✕ REMOVE

The Pioneer electronics

DWX2305 mainboard, the CD drive + optical pickup, the audio daughterboard, the rear output board and the OEM power supply.

Out: a Pioneer MCU.
In: an ESP32-S3.

The DWX2305 mainboard comes out. A single **ESP32-S3 DevKitC-1** drops into its footprint and speaks class-compliant USB-MIDI natively via TinyUSB.

- → **Native USB-OTG** — class-compliant MIDI, no adapter.
- → **PCNT + dual ADC** — hardware jog quadrature.
- → **WiFi / BLE + PSRAM** — room for Link & a future jog display.
- → **Current silicon** — cheaper than the Teensy 3.6 of earlier builds.

↑ The OEM DWX2305 MAIN — Pioneer MCU + Xilinx Spartan FPGA. Removed in v0.1.

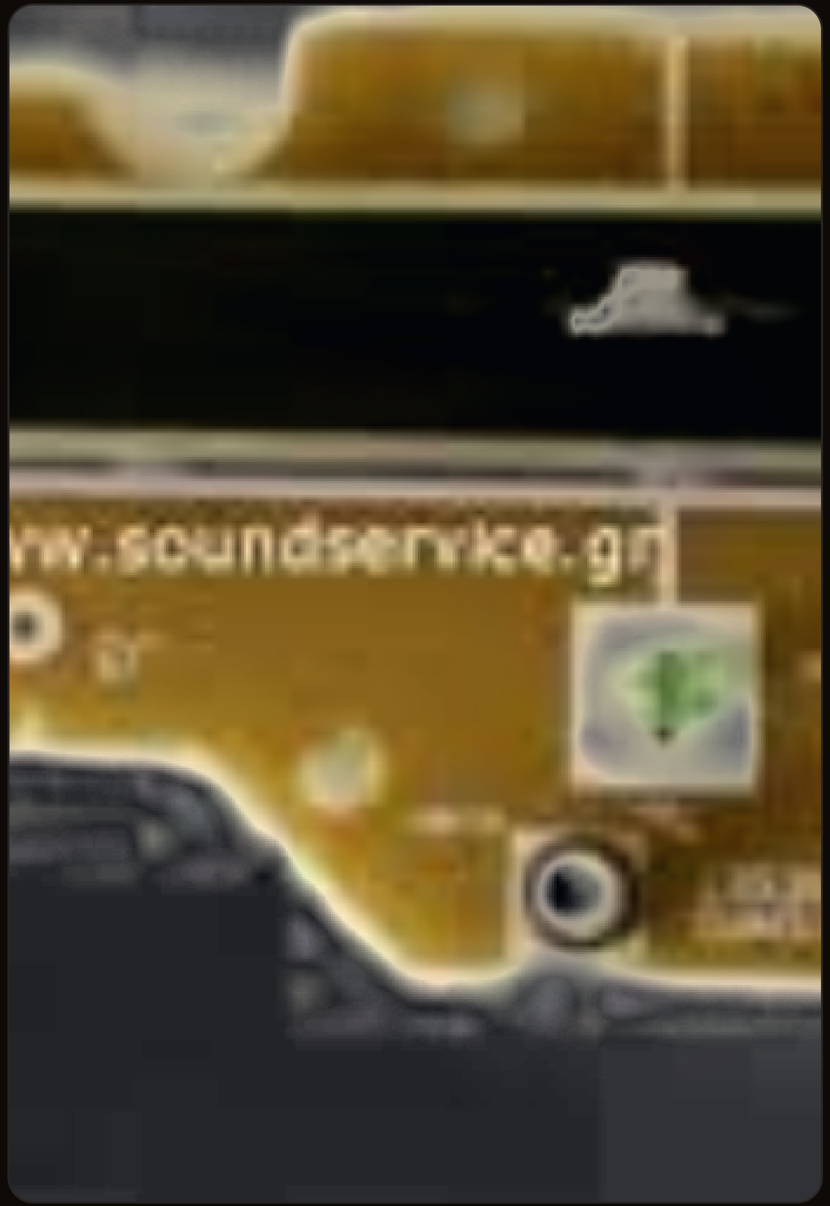
THE TEARDOWN

Every board, accounted for

Silkscreen part numbers read off the open unit map cleanly to a keep / remove decision.

▲ KEEP

DWS1365 / DWX2306 KSWB
button PCBs → mux

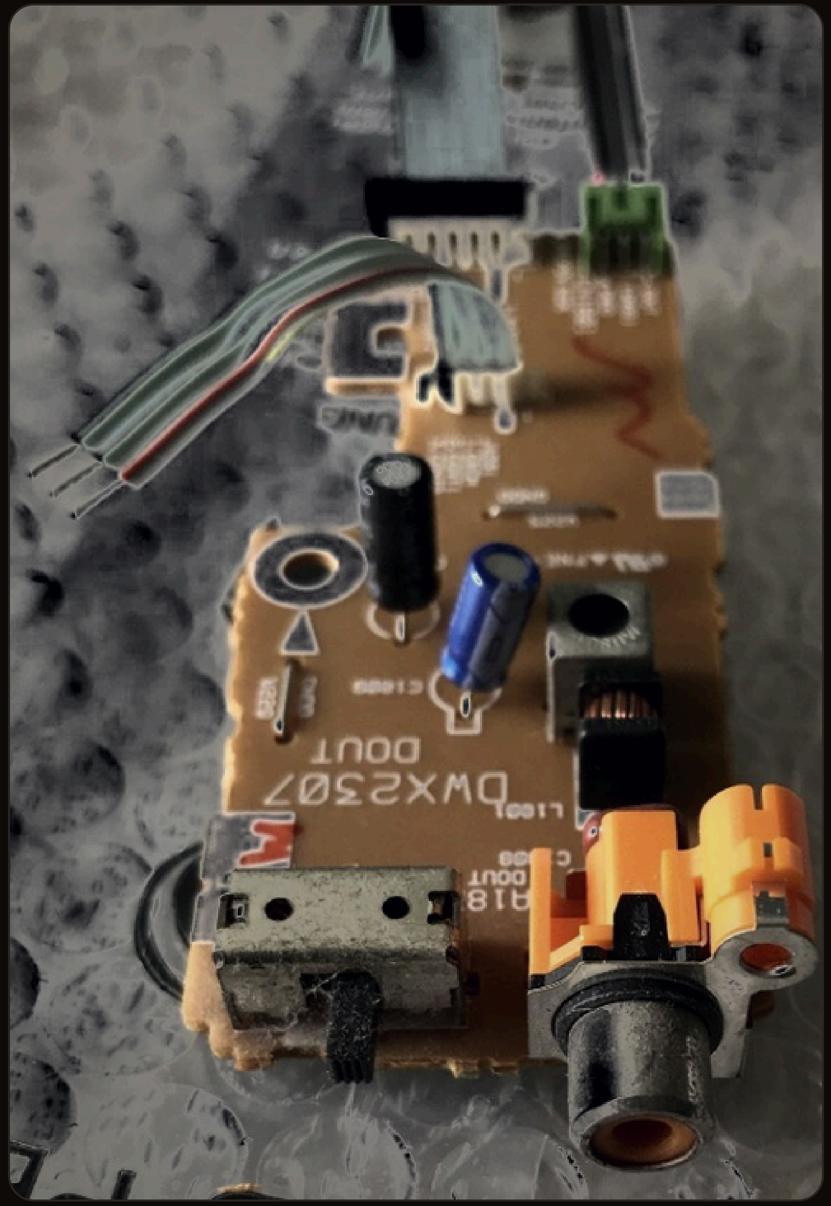


▲ KEEP

Pitch fader · 100 mm
wiper → S3 ADC

✕ REMOVE

DWX2305-D ABB
audio daughter board



✕ REMOVE

DWX2387 OUT
rear RCA + AC input

Every control, re-routed to the S3

A small carrier PCB sits in the old mainboard footprint and turns raw OEM signals into S3-friendly ones.

Jog encoder
CH A / CH B optical

Jog touch
sheet sense

Pitch fader
100 mm linear pot

~20 buttons
incl. Hot Cue A/B/C + Loop

CUE / PLAY / LOOP LEDs
12 V illuminated rings

→ → →

CARRIER PCB

TXS0108E
5 V ↔ 3.3 V level shift

2× 74HC4067
16→1 button muxes A + B

IRLZ44N / AO3400
low-side LED switching

MT3608 boost
USB 5 V → 12 V rail

→ → →

PCNT
GPIO 4 / 5 — quadrature

Touch in
GPIO 6

ADC1_CH0
GPIO 1 — ⚠ re-fed at 3.3 V

Mux SIG + select
GPIO 8–13

MOSFET gates
GPIO 14 / 19 / 20 / 21

The jog wheel earns its keep

Three signals make the platter sing: an **optical encoder** at ~135 frames per revolution, a **touch sheet** for vinyl-mode scratch, and the blue **“vinyl” indicator LED** glowing at its hub.

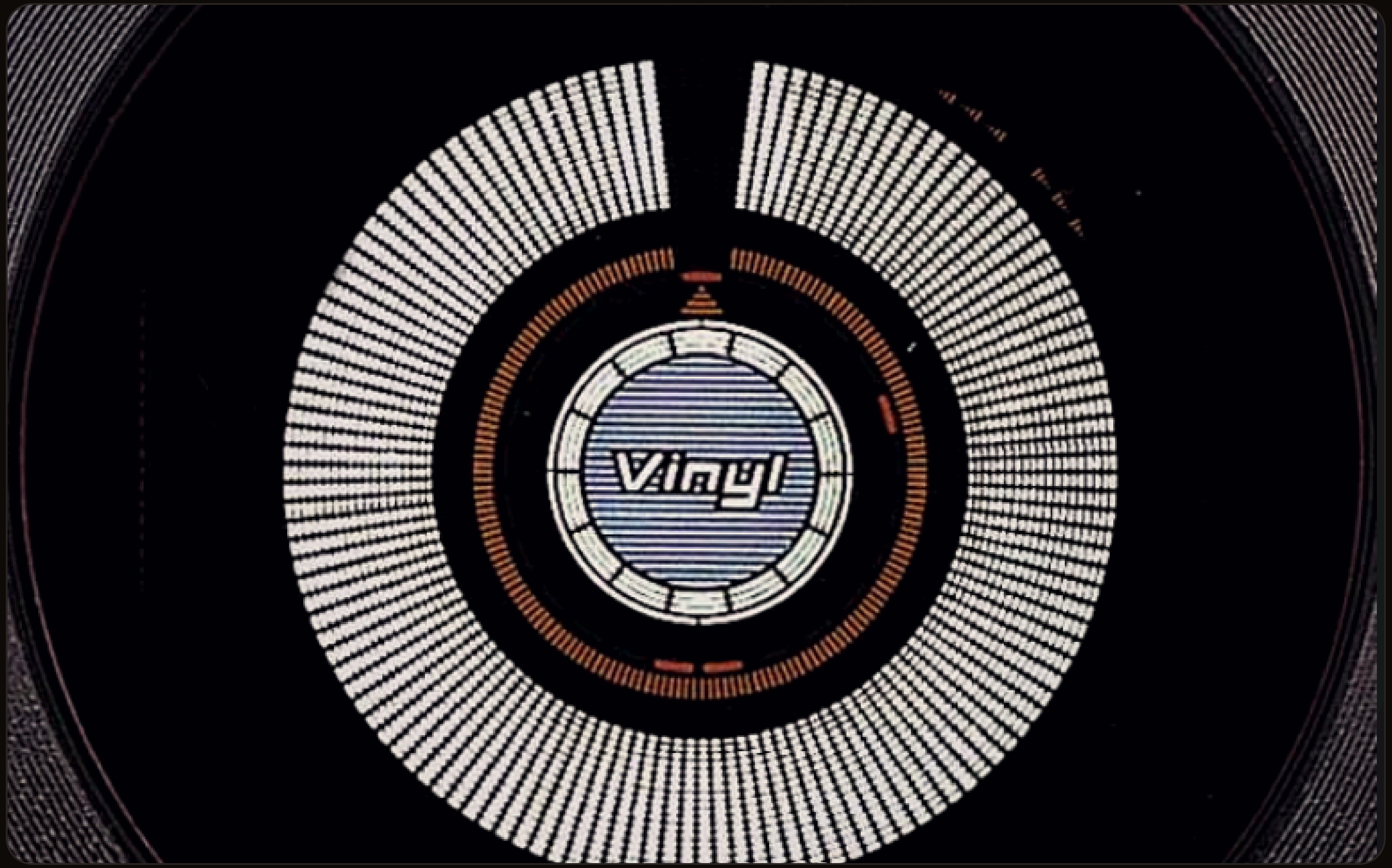
- CH A / CH B → PCNT
- touch → GPIO 6
- vinyl LED → MOSFET

The VFD stays dark — for now.

No one has driven the CDJ-1000's OEM vacuum-fluorescent display from a custom MCU. For v0.1 the glass goes dark — Traktor on the laptop carries the rich UI — and only the plain status LEDs are re-used.

⚠ THE CAPTURE WINDOW

Before the mainboard comes out, logic-analyze the jog-centre position byte off the MAIN↔JFLB ribbon. **That bus dies the moment the brain is removed** — so the capture keeps the v0.2 jog-display option alive.



v0.1 · LEDs only

µPD16306B VFD driver left unpowered.

v0.2 · Path D

Port djgreeb's MK3 STM32 panel to MK2.

v0.1 scope — locked

- ✓ **ESP32-S3 is the sole brain** — no STM32, no big screen.
- ✓ **VFD driver identified** — NEC μPD16306B on JFLB DWG1568.
- ✓ **Board catalog confirmed** from silkscreen part numbers.
- ✓ **Big-screen path documented** (Path D) and deferred to v0.2+.

Open at the bench

- ○ Confirm MK2 jog encoder voltage & PPR from manual `RRV2802`.
- ○ Jog-touch sensor type — capacitive vs pressure sheet.
- ○ Trace button connectors → final 4067 channel map (maybe a 3rd mux).
- ○ **Capture the jog-centre byte** before disassembly.
- ○ Pick firmware path — ESP-IDF vs Arduino-ESP32 TinyUSB.

PIONEER CDJ-1000MK2 → ESP32-S3

Old iron. New brain. One-of-a-kind deck.

repo github.com/gbroeckling/cdj1000 license GPL-3.0

STANDING ON THE SHOULDERS OF

- djgreeb · MK3 new life
- spectran · CDJ-100S adapter
- pestrela · dj_maps
- Lee Smith · DJLegionUK